

OBJECT ORIENTED PROGRAMMING

EXPERIMENT - 1

INTRODUCTION TO JAVA ENVIRONMENT

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BATCH : 2

Ques-1 Explore & understand the role of JDK, JRE, JVM.

Ans: The main difference b/w JDK, JRE and JVM plays a very important role in understanding how JAVA plays.

The main difference between JDK, JRE & JVM is :-

JDK: Java development kit is a software development environment used for developing Java applications.

JRE: JRE stands for Java runtime environment and it provides an environment to run only the Java program onto the system.

JVM: JVM stands for JAVA virtual machine and is responsible for executing the Java program.

<u>Aspect:</u>	<u>JDK</u>	<u>JRE</u>	<u>Jvm</u>
Purpose	used to develop Java applications	use to run Java applications	Responsible for running Java code
Platform dependency	Platform development	Platform dependent	Platform independent
Use case	It include development tools like compiler +	Running a Java application	convert byte-code into

Ques: Install latest available JDK and verify the Java Environment.

Java(TM) SE Development Kit 23.0.2 (64-bit) Successfully Installed

Click Next Steps to access tutorials, API documentation, developer guides, release notes and more to help you get started with the JDK.

Next Steps

Close

```
C:\Users\VAANI JINDAL>java --version
java 23.0.2 2025-01-21
Java(TM) SE Runtime Environment (build 23.0.2+7-58)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.2+7-58, mixed mode, sharing)

C:\Users\VAANI JINDAL>javac --version
javac 23.0.2

C:\Users\VAANI JINDAL>
```

Ques 3 Create a sample HelloWorld program using simple text editor (eg Notepad) and show the steps to compile and execute the program using command prompt.

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

```
C:\Users\VAANI JINDAL>java --version  
java 23.0.2 2025-01-21  
Java(TM) SE Runtime Environment (build 23.0.2+7-58)  
Java HotSpot(TM) 64-Bit Server VM (build 23.0.2+7-58, mixed mode, sharing)  
  
C:\Users\VAANI JINDAL>javac --version  
javac 23.0.2  
  
C:\Users\VAANI JINDAL>cd C:\Users\VAANI JINDAL\OneDrive\Desktop  
C:\Users\VAANI JINDAL\OneDrive\Desktop>javac HelloWorld.java  
C:\Users\VAANI JINDAL\OneDrive\Desktop>java HelloWorld  
Hello, World!  
C:\Users\VAANI JINDAL\OneDrive\Desktop>
```

Commands:-

cd → change directory to the project folder
javac → compile the java file
java → Run the Java program

ps:
step 1:

Write the Java program using notepad
Save the file using .java extension.

step 2:

Compile the Java program using command prompt
Navigate to the folder where you saved HelloWorld
eg. cd path.

step 3:

Compile the java program using javac command
eg. javac HelloWorld.java

step 4:

Run the compiled Java programs using the Java
command

eg. java HelloWorld

Output :- Hello, world!

Ques 4

Display your name and complete address in different lines

```
package basics;

public class ques4 {

    public static void main(String[] args) {
        System.out.println("Vaani Jindal");
        System.out.println("UPES, Dehradun");
        System.out.println("Uttarakhand, India");
    }
}
```

```
<terminated> ques4 [Java Application] C:\Users\VAANI JINDAL\.p2\pool\p
Vaani Jindal
UPES, Dehradun
Uttarakhand, India
```